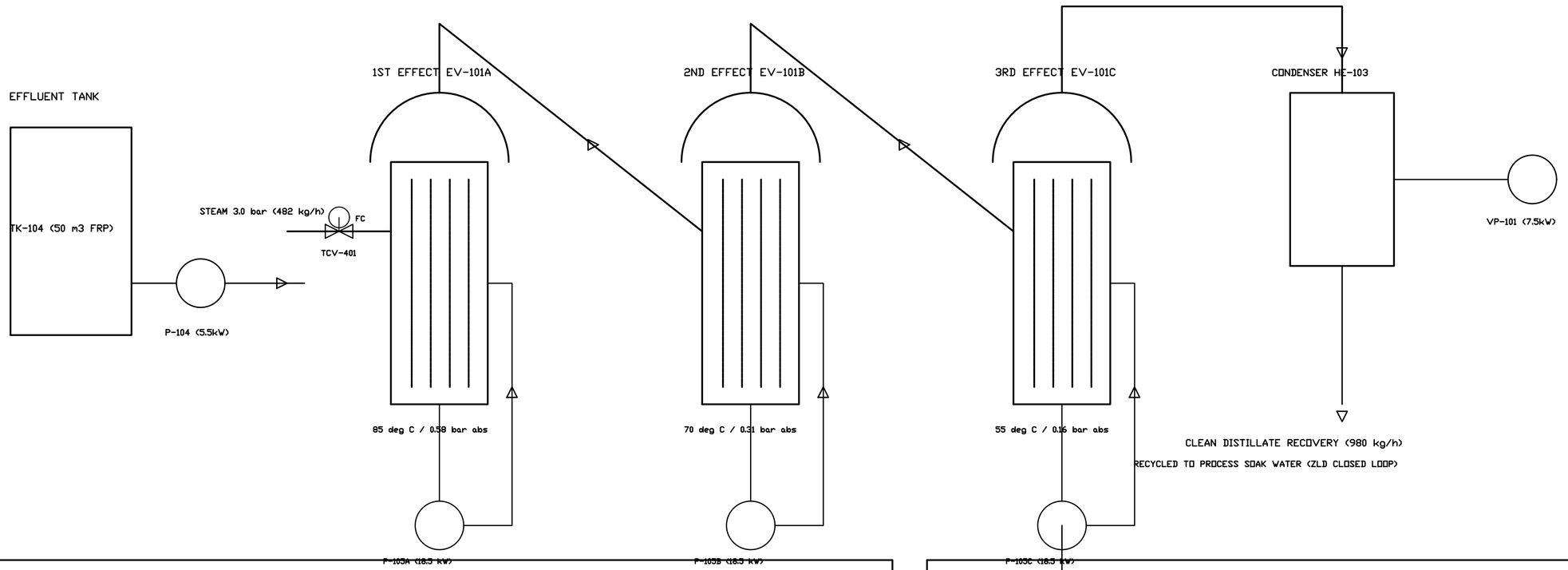


ZLD TRIPLE-EFFECT THERMAL PERFORMANCE & MASS BALANCE SCHEDULE

EFFECT / UNIT	OP. TEMP	ABS PRESS	EVAP RATE	FEED / OUTLET CONC	HEAT DUTY	SPECIFIC STEAM
1st Effect (EV-101A)	85.0 deg C	0.58 bar a	340.0 kg/h	0.63% -> 0.95% wt	225.4 kW	0.42 kg steam / kg H2O
2nd Effect (EV-101B)	70.0 deg C	0.31 bar a	325.0 kg/h	0.95% -> 1.88% wt	215.1 kW	Re-uses 1st effect vapor
3rd Effect (EV-101C)	55.0 deg C	0.16 bar a	315.0 kg/h	1.88% -> 40.0% wt	208.7 kW	Re-uses 2nd effect vapor
Condenser (HE-103)	38.0 deg C	0.14 bar a	315.0 kg/h	Distillate TDS < 25 ppm	212.0 kW	Cooling water 32 -> 38 deg C
OVERALL ZLD SYSTEM	Net dT=30C	PR = 2.38	980.0 kg/h	96.2% Water Recovery	649.2 kW	0.00 m3/day Liquid Effluent

ZLD EVAPORATOR INSTRUMENTATION & ISA-5.1 CONTROL LOOPS

LOOP TAG	MEASURED PARAMETER	PRIMARY SENSOR	CONTROL ELEMENT	FAIL ACTION	NORMAL SETPOINT
TIC-401	1st Effect Calandria Temp	Dual RTD PT100 (3-wire)	TCV-401 Steam Globe Valve	Fail Closed (FC)	85.0 deg C +/- 0.5 deg C
LIC-401	1st Effect Sump Level	HART DP Level Transmitter	LCV-401 Inter-Stage Valve	Fail Closed (FC)	450 mm (50% Sump Height)
LIC-402	2nd Effect Sump Level	HART DP Level Transmitter	LCV-402 Inter-Stage Valve	Fail Closed (FC)	450 mm (50% Sump Height)
DIC-403	3rd Effect Slurry Density	Coriolis Mass/Density Meter	LCV-403 Concentrate Bleed	Fail Closed (FC)	1.185 g/cm3 (40 wt% solids)
PIC-404	Vacuum Header Pressure	Capacitance Manometer	PCV-404 Vacuum Bleed Valve	Fail Open (FO)	0.140 bar a +/- 0.005 bar

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

P&ID - TRIPLE-EFFECT ZLD WASTEWATER EVAPORATOR
DISCIPLINE: PIPING | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-010	REV: V8.0	SHEET: 10 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ